

Mechanika Tuhých a Poddajných Telies

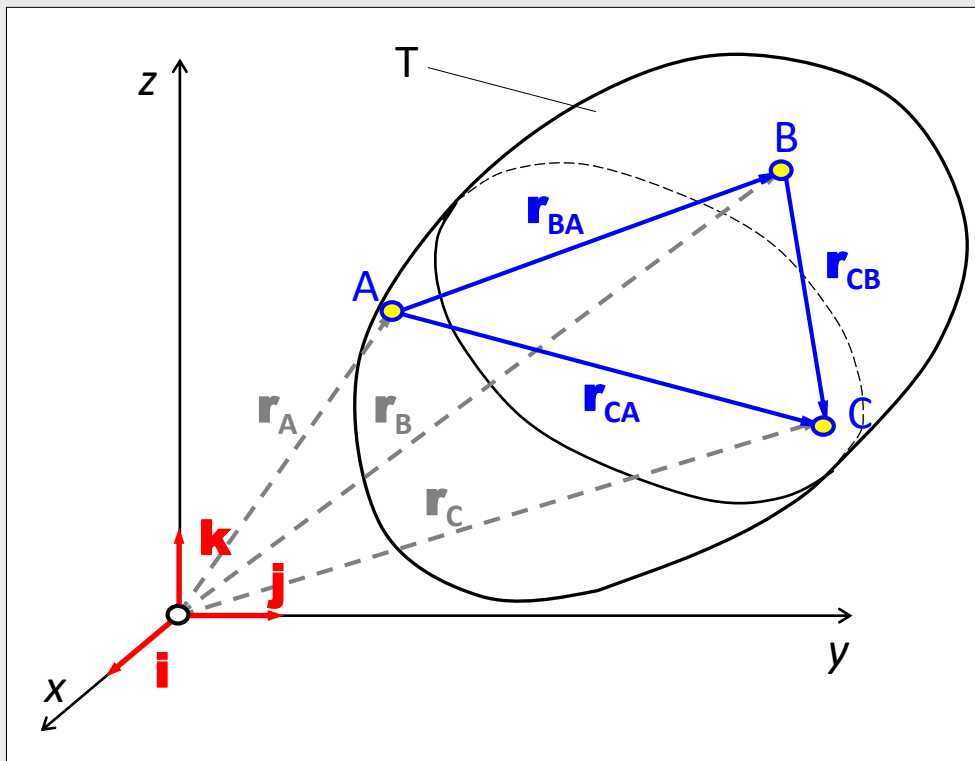
(MTPT)

P-06

Kinematika tuhého telesa

Tuhé teleso - sústava hmotných bodov (nekonečný počet) - ich vzájomná poloha pri ľubovoľnom premiestnení, t.j. pohybe nemení.

Matematická formulácia tuhého telesa:



$$A, B, C \in T$$

$$A \neq B \wedge B \neq C \wedge C \neq A$$

$\wedge$ - „a súčasne“

$$\mathbf{r}_{BA} = \mathbf{r}_B - \mathbf{r}_A$$

$$\mathbf{r}_{CA} = \mathbf{r}_C - \mathbf{r}_A$$

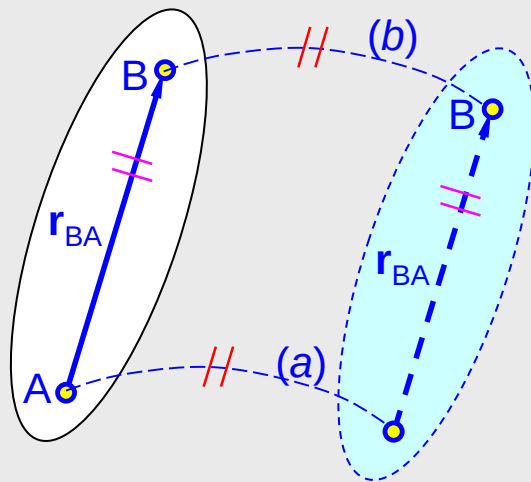
$$\mathbf{r}_{CB} = \mathbf{r}_C - \mathbf{r}_B$$

$$|\mathbf{r}_{BA}| = \text{konšt.} \wedge |\mathbf{r}_{CA}| = \text{konšt.} \wedge |\mathbf{r}_{CB}| = \text{konšt.}$$

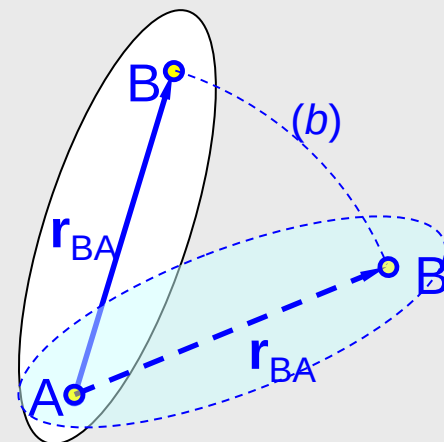
Pohyb tuhého tělesa

• v rovine :

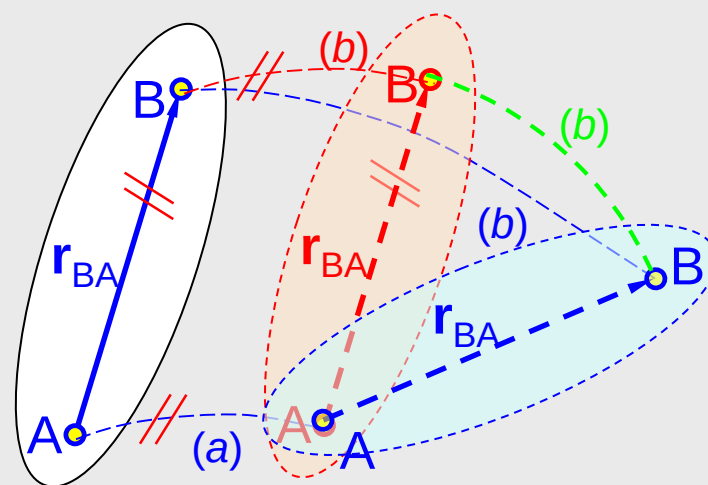
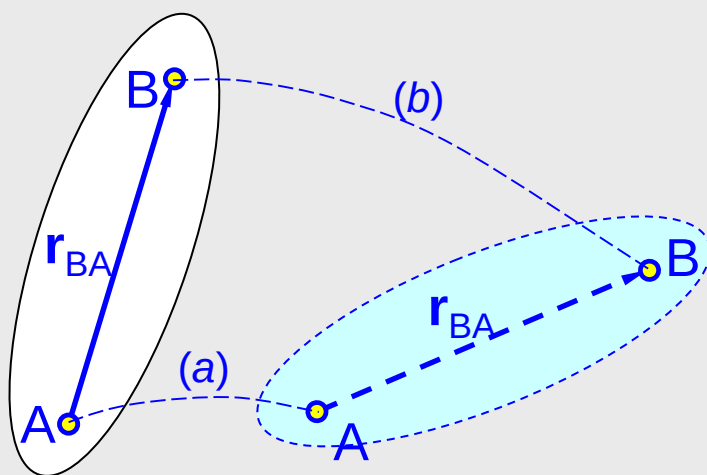
- posuvný pohyb



- rotačný pohyb



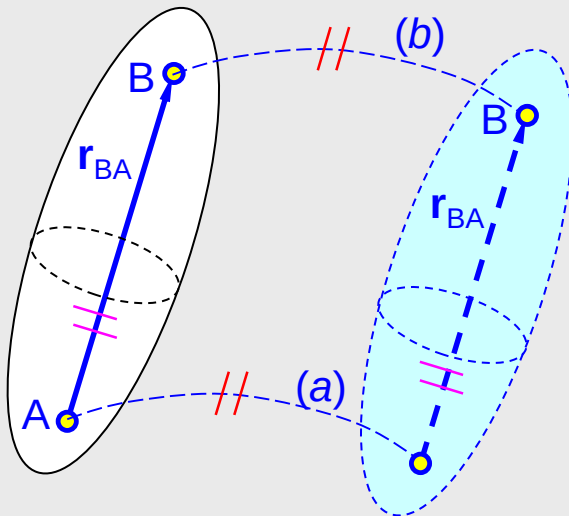
- všeobecný pohyb



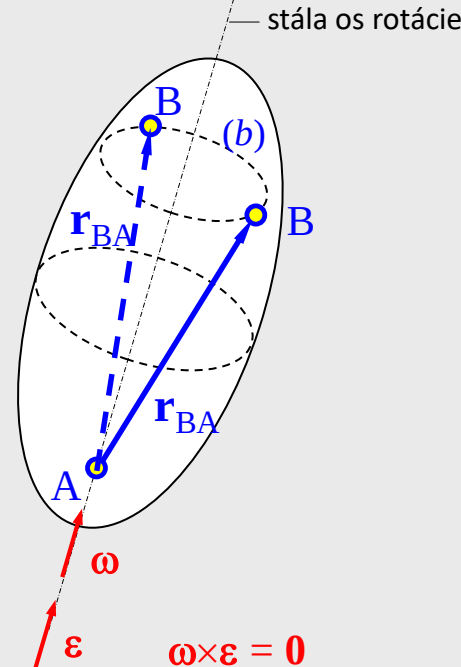
Pohyb tuhého telesa

- v priestore :

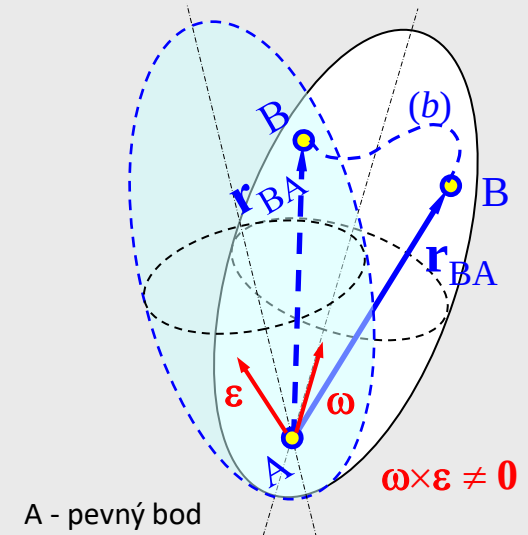
- posuvný pohyb



- rotačný pohyb



- sférický pohyb



- všeobecný pohyb

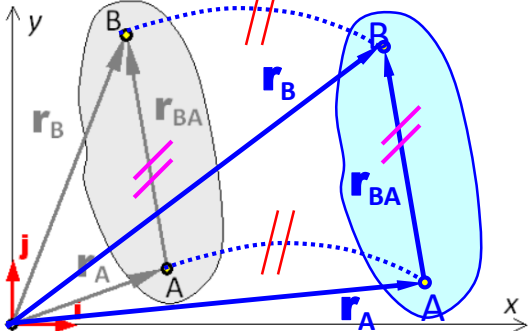
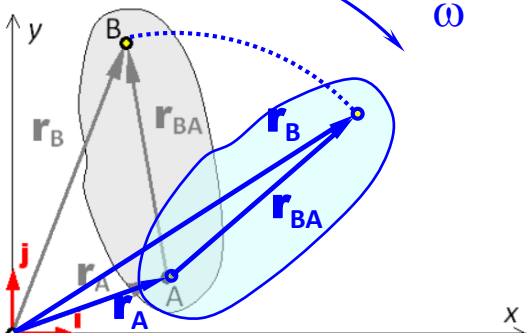
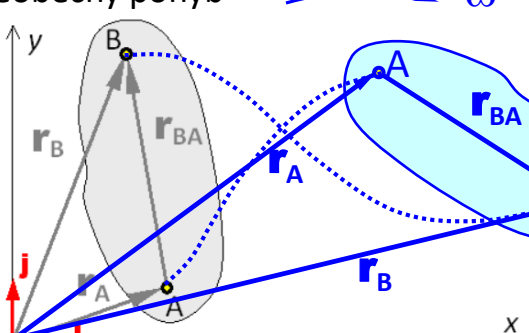
všeobecný pohyb = posuvný pohyb + rotačný pohyb

všeobecný pohyb = posuvný pohyb + sférický pohyb

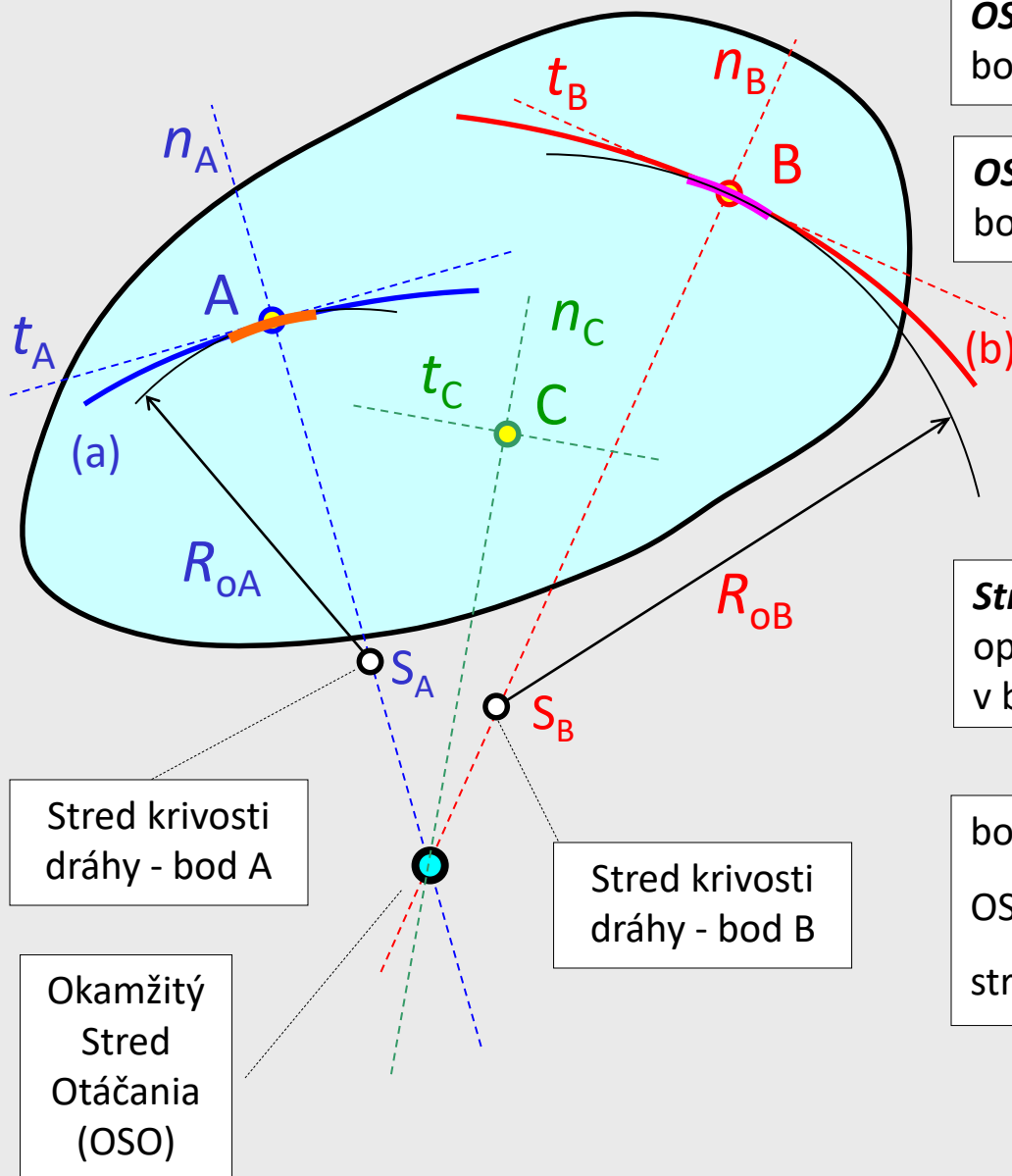
KINEMATIKA

k - konštanta

m - mení sa

MODEL	$\mathbf{r}_A$	$ \mathbf{r}_A $	$\mathbf{r}_B$	$ \mathbf{r}_B $	$\mathbf{r}_{BA}$	$ \mathbf{r}_{BA} $	ROVNICE
posuvný pohyb 	m	m	m	m	k	k	$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{BA} \quad \cdot$ $\mathbf{v}_B = \mathbf{v}_A + \mathbf{0} = \mathbf{v}_A \quad \cdot$ $\mathbf{a}_B = \mathbf{a}_A + \mathbf{0} = \mathbf{a}_A$
rotačný pohyb 	k	k	m	m	m	k	$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{BA} \quad \cdot$ $\mathbf{v}_B = \mathbf{0} + \mathbf{v}_{BA} = \boldsymbol{\omega} \times \mathbf{r}_{BA} \quad \cdot$ $\mathbf{a}_B = \mathbf{0} + \mathbf{a}_{BA} = \boldsymbol{\epsilon} \times \mathbf{r}_{BA} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{BA})$
všeobecný pohyb 	m	m	m	m	m	k	$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{BA} \quad \cdot$ $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{BA} = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{BA} \quad \cdot$ $\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{BA} = \mathbf{a}_A + \boldsymbol{\epsilon} \times \mathbf{r}_{BA} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{BA})$

Všeobecný pohyb telesa



OSO - priesečník normál k trajektóriám (dráham) bodov daného telesa v danom okamihu

OSO - v danom okamihu je spoločný pre všetky body telesa

$$v_{OSO} = 0$$

Stred krivosti dráhy - bod na normále - v ňom opísaná kružnica s polomerom krivosti R_o nahradí v blízkom okolí bodu telesa jeho dráhu.

Stred krivosti dráhy - bod A

Stred krivosti dráhy - bod B

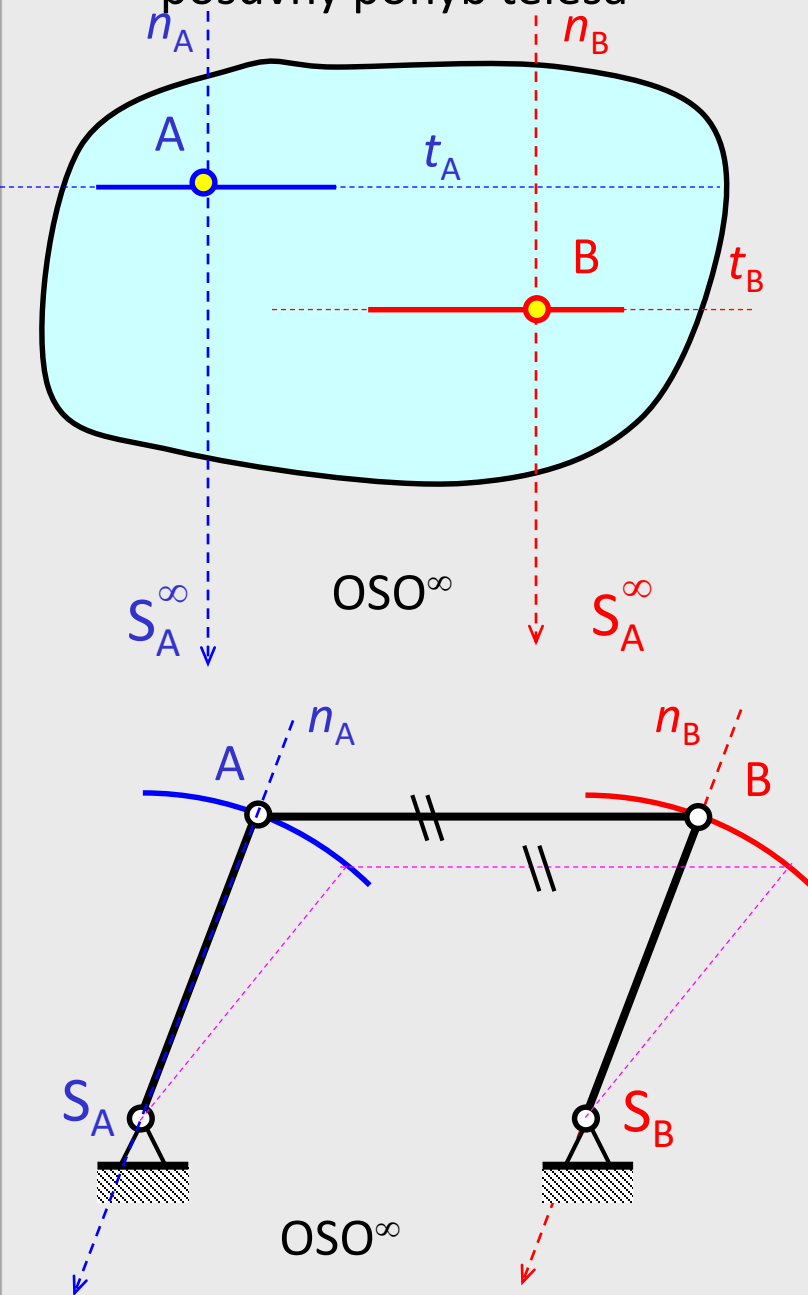
Okamžitý
Stred
Otáčania
(OSO)

bod telesa (napr. A)
OSO
stred krivosti (S_A)

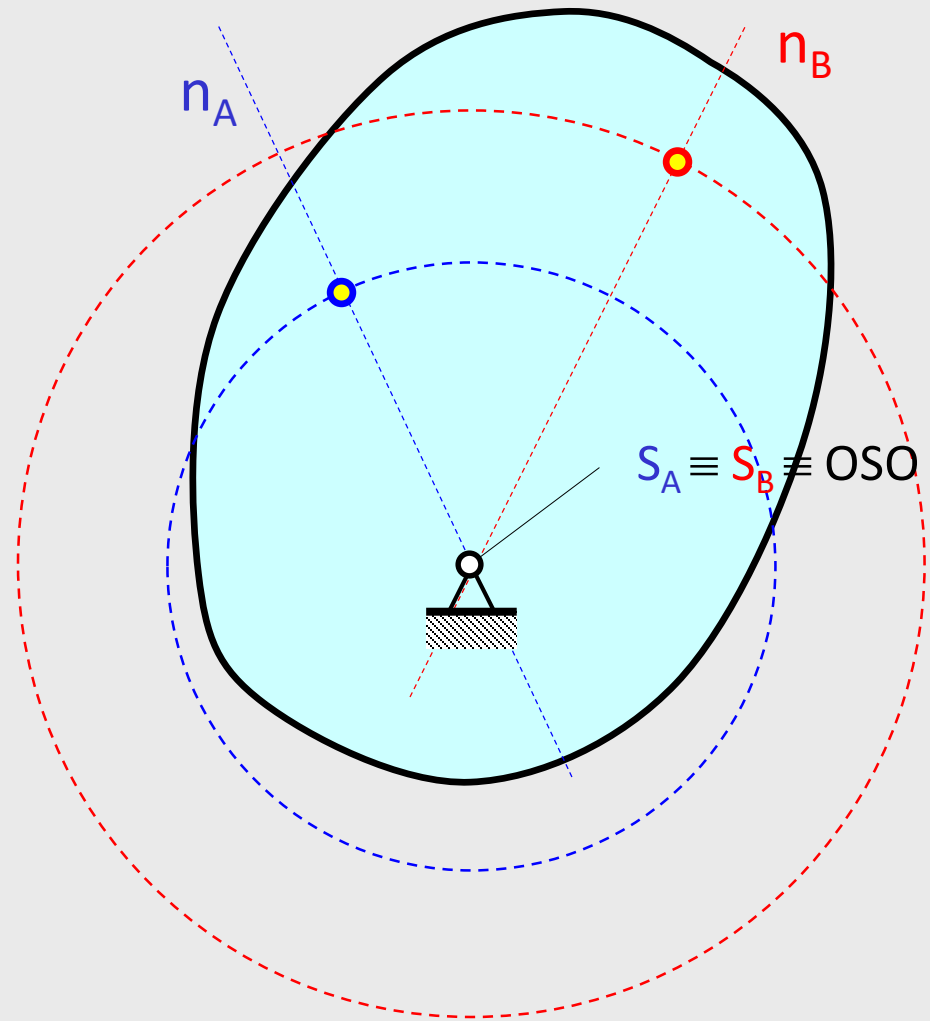
} ležia na normále n_A

KINEMATIKA

posuvný pohyb telesa

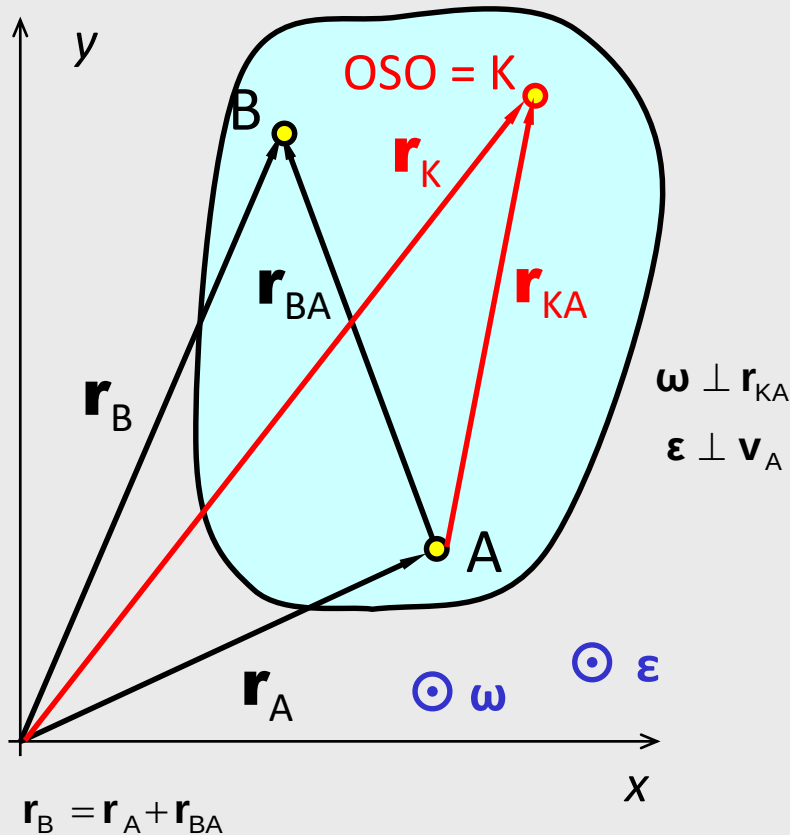


rotačný pohyb telesa



KINEMATIKA

Polohový vektor OSO



$$\mathbf{r}_K = \mathbf{r}_A + \mathbf{r}_{KA}$$

$$\mathbf{v}_K = \mathbf{v}_A + \mathbf{v}_{KA} = \mathbf{v}_A + \omega \times \mathbf{r}_{KA}$$

$$\mathbf{v}_K = 0$$

$$0 = \mathbf{v}_A + \omega \times \mathbf{r}_{KA} \quad / \omega \times$$

$$-\mathbf{v}_A = \omega \times \mathbf{r}_{KA}$$

$$0 = \omega \times \mathbf{v}_A + \omega \times (\omega \times \mathbf{r}_{KA})$$

$$0 = \omega \times \mathbf{v}_A + \omega (\omega \cdot \mathbf{r}_{KA}) - \mathbf{r}_{KA} (\omega \cdot \omega)$$

$$0 = \omega \times \mathbf{v}_A - \omega^2 \mathbf{r}_{KA}$$

$$\omega \times \mathbf{v}_A = \omega^2 \mathbf{r}_{KA}$$

$$\mathbf{r}_{KA} = \frac{\omega \times \mathbf{v}_A}{\omega^2}$$

$$\mathbf{a}_K = \mathbf{a}_A + \mathbf{a}_{KA} = \mathbf{a}_A + \epsilon \times \mathbf{r}_{KA} + \omega \times (\omega \times \mathbf{r}_{KA})$$

$$\mathbf{a}_K = \mathbf{a}_A + \epsilon \times \mathbf{r}_{KA} - \omega \times \mathbf{v}_A$$

$$\mathbf{a}_K = \mathbf{a}_A + \epsilon \times \mathbf{r}_{KA} - \omega \times \mathbf{v}_A = \mathbf{a}_A - \frac{\epsilon}{\omega} \mathbf{v}_A - \omega \times \mathbf{v}_A$$

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{BA} = \mathbf{a}_A + \epsilon \times \mathbf{r}_{BA} + \omega \times (\omega \times \mathbf{r}_{BA})$$

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$$

$$\epsilon \times \mathbf{r}_{KA} = \frac{1}{\omega^2} [\epsilon \times (\omega \times \mathbf{v}_A)] = \frac{1}{\omega^2} [\omega (\epsilon \cdot \mathbf{v}_A) - \mathbf{v}_A (\epsilon \cdot \omega)] = -\frac{\epsilon \omega}{\omega^2} \mathbf{v}_A = -\frac{\epsilon}{\omega} \mathbf{v}_A$$